FILTER DESIGN TEST CASE

High-Pass Filter - Band 2 (1900 MHz) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Author:	Dr. Elena Vasquez
Reviewer:	Dr. Anya Krishnamurthy
Design Center:	Cedar Rapids Lab
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Priority:	P1 - Critical

1. OBJECTIVE

This document describes the design, simulation, and characterization of a High-Pass Filter filter targeting Band 2 (1900 MHz) for Automotive V2X Communication. The filter is designed on ScAlN on Si substrate with a target center frequency of 1347.3 MHz and bandwidth of 173.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: High-Pass Filter
- Frequency Band: Band 2 (1900 MHz)
- Application: Automotive V2X Communication
- Substrate: ScAlN on Si
- Package: CSP (Chip Scale Package)
- Design Tool: PathWave EM Design
- Design Center: Cedar Rapids Lab

This specification applies to all variants of the High-Pass Filter design targeting Band 2 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 1347.3 MHz
Bandwidth (3 dB): 173.1 MHz
Insertion Loss Target: <= 2.8 dB
Return Loss Target: >= 9.0 dB
Out-of-Band Rejection: >= 26.0 dB
Isolation (Duplexer): >= 29.0 dB
Q Factor: 6512
Temperature Coefficient:-10.8 ppm/°C
Die Size: 2.8 x 1.1 mm
Wafer Size: 8-inch
Number of Resonators: 4
Electrode Material: Al

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open PathWave EM Design and load the High-Pass Filter template project.
2. Define the substrate stack: ScAlN on Si with specified layer thicknesses.
3. Set design targets: center freq 1347.3 MHz, BW 173.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.8 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, ScAlN on Si).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 4042 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.8 dB
- Return loss in passband shall be >= 9.0 dB
- Out-of-band rejection at +/- 260 MHz offset >= 26.0 dB
- Group delay variation across passband shall be <= 8 ns
- Temperature drift over -40°C to +85°C shall be <= 1.8 MHz
- Power handling shall be >= +32 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 88%

6. TEST RESULTS

Measurement Date: 2024-09-19
Devices Tested: 46
Insertion Loss (meas): 2.5 dB
Return Loss (meas): 10.6 dB
Rejection (meas): 28.8 dB
Isolation (meas): 33.8 dB
Q Factor (meas): 6512
Group Delay Variation: 6.3 ns
Power Handling: +34 dBm
Die Yield: 84.6%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

a) The High-Pass Filter design demonstrated below-target performance for Band 2 (1900 MHz).
b) Insertion loss is marginally above target; electrode thickness optimization recommended.
c) Rejection at near-band offset requires additional resonator tuning.
d) Temperature stability was within specification across full range.
e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

Author:	Dr. Elena Vasquez	Date: 2024-09-19
Reviewer:	Dr. Anya Krishnamurthy	Date: 2024-09-26
Approver:	Dr. James Liu	Date: 2024-10-03

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